

## Exercise 08 - Stack & Queue Problems

For each problem, create a cpp file named "element0 $n$ .cpp" where  $n$  is the problem number. Each source code must use a *Stack* or *Queue* object.

### Problems:

1. Define the string function named RCD() whose header is

```
string RCD(string str)
```

It returns a reduced string of  $str$  that contains no consecutive duplicates and maintains the order of characters of  $str$ .

#### Example:

For instance, the caller RCD("cabbacdbc") will return "dbc".

Afterward, within the main function, verify the sample from the example.

- 2.

3. Define the void function named BC() whose header is

```
void BC(size_t n)
```

that displays the count to  $n$  starting from 1 in binary numbers, each on a separate line if  $n$  is positive.

#### Example:

For instance, if  $n = 7$ , the function will display "1\n10\n11\n100\n101\n110\n111".

Afterward, within the main function, verify the sample from the example.